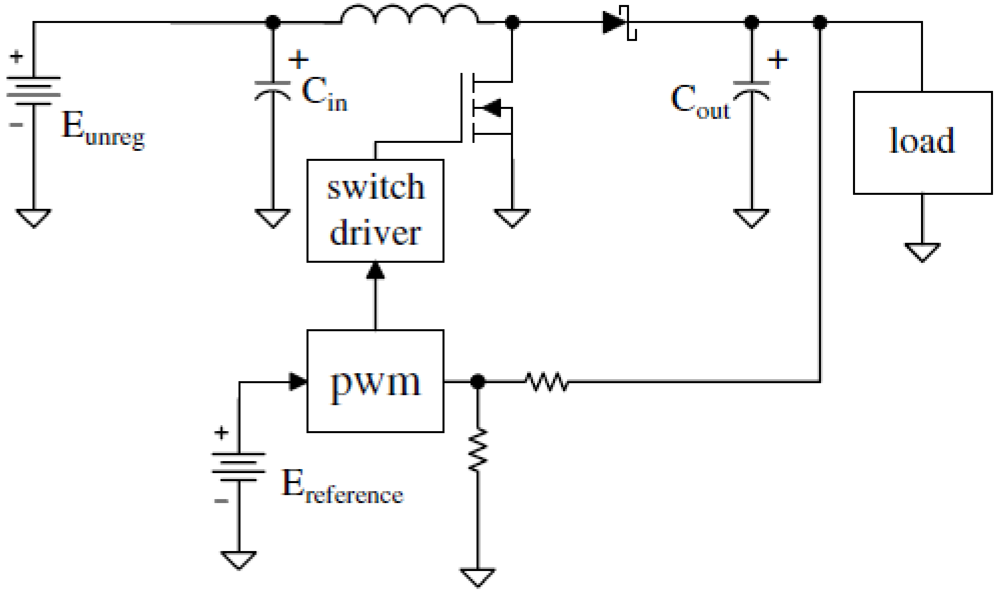
POWER ELECTRONICS

Switching Power Supplies

Part 2

Boost Regulators/Converters

- Low side switch instead of high side
- When transistor is on, diode is off and vice versa
- When transistor is on: $v_L = E_{unreg} = L\frac{\Delta i}{\Delta t}$
- Analysis vs Design scenario
- Boost regulators are 90% efficient
- $P_{load} = 0.9P_{in}$



Boost Regulators/Converters

- Average inductor current (using power equation)
- Inductor is directly connected to input so inductor

current is also the input current

- $I_{P\ L} = I_{dc\ L} + \frac{\Delta i}{2}$
- Average transistor current, rms value?
- Average diode current, rms value?

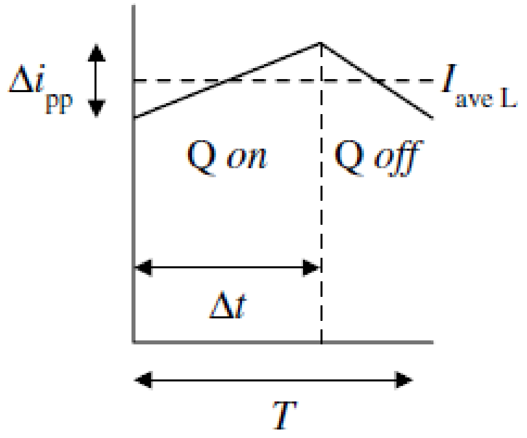


Figure Inductor current

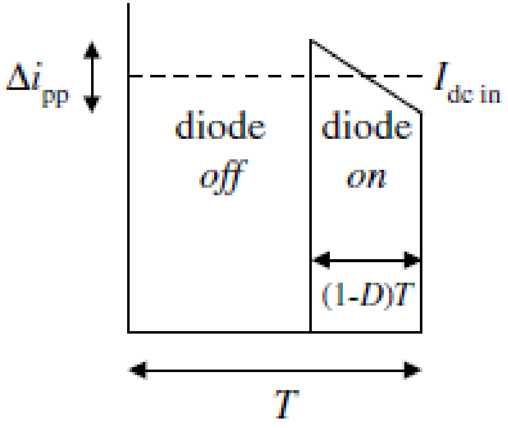


Figure Diode current

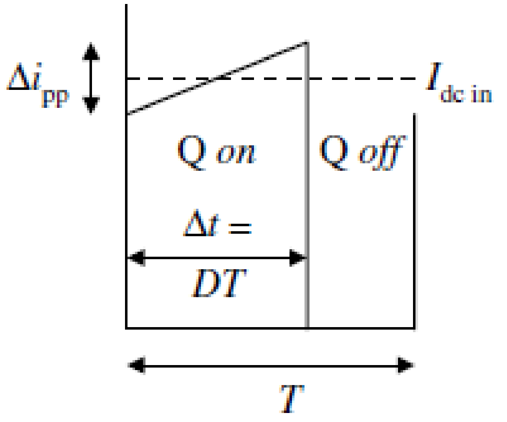
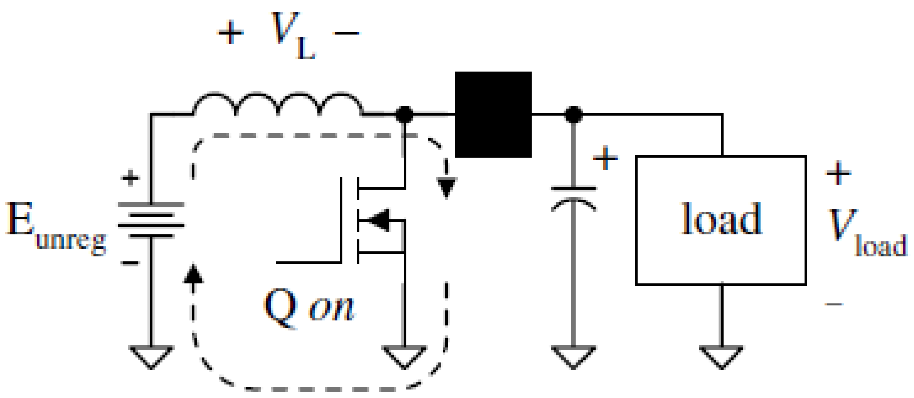


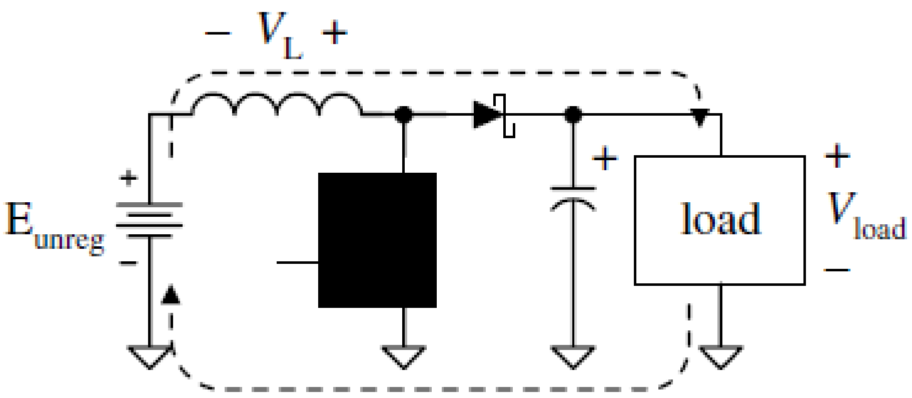
Figure Transistor current

Boost Regulators/Converters

- How output is higher than the input?
- Inductor voltage in both parts?
- $v_{load} = E_{unreg} + v_L$
- Load current and the output capacitor relation?
- Keep $\Delta v_{load} < 20\ mV_{pp}$



(a) Inductor charges with Q *on*



(b) Inductor flies back with Q *off*

Example

Design a boost regulator that converts an input voltage of 12 V dc to a load voltage of 22 V dc at 0.7 A dc. Use a controller frequency of 100 kHz. Assume that the transistor has an on resistance of 0.4 Ω and the diode drops 0.5 V when it is conducting.

Calculate the power and current that must be provided to the regulator, the inductor's inductance and current ratings, the power that the transistor and the diode must dissipate, and the value of the output capacitor.

Example 2

Design a boost regulator that converts an input voltage of 22 V dc at 0.7 A dc to a load voltage of 56 V dc. Use a controller frequency of 100 kHz. Assume that the transistor has an on resistance of 0.4 Ω and the diode drops 0.5 V when it is conducting.

Calculate the power and current that must be provided to the regulator, the inductor's inductance and current ratings, the power that the transistor and the diode must dissipate, and the value of the output capacitor.

LM2585 Boost Regulator

- Buck regulator IC by National Semiconductor "Simple Switcher" Series.
- $V_{dc\ load} = 1.23V_{dc}\left(1 + \frac{R_f}{R_i}\right)$ where $1\ \text{k}\Omega < R_i < 5\ \text{k}\Omega$
- $P_Q = I_{rms\ Q}^2 R_{on\ Q}$ where $I_{rms\ Q} = \sqrt{D} I_{dc\ load}$ and $R_{on\ Q} = 0.15\ \Omega$
- $P_{IC} = \frac{1}{50} D I_{dc\ in} E_{unreg}$
- Total Power dissipation $P_{LM2585} = P_Q + P_{IC}$

Example 3

- Design a boost regulator that converts an input voltage of 12 V dc to a load voltage of 22 V dc at 0.7 A dc using LM2585. Calculate the feedback resistors, power, and current that must be provided to the regulator, the power that the IC must dissipate, and the junction temperature assuming $T_A = 80°\text{C}$ and that there is no heat sink.

LM2585 thermal parameters

$T_{J\text{ upper limit}} = 110°\text{C}$

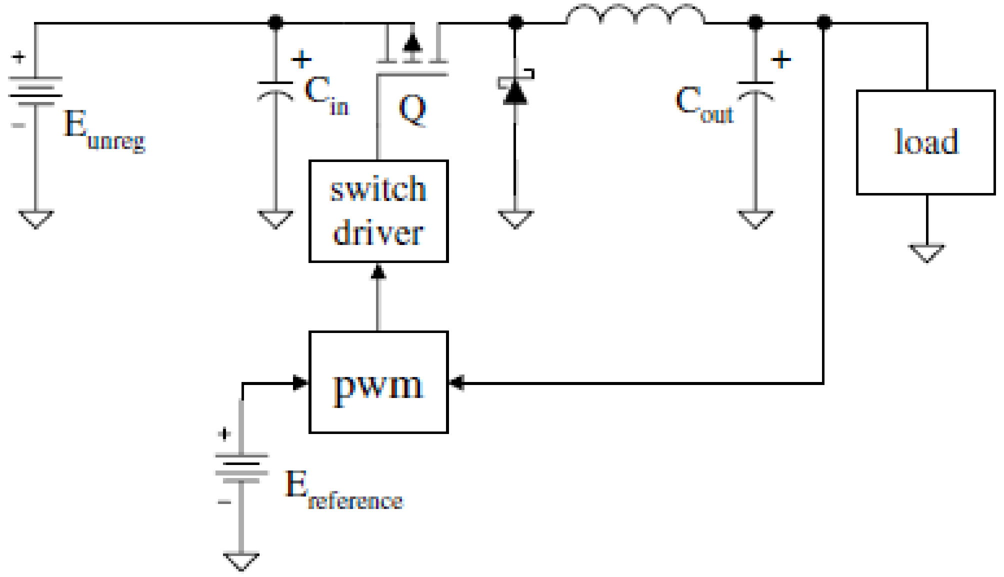
$\Theta_{JA\text{ bare case}} = 65°\text{C/W}$

$\Theta_{JC} = 2°\text{C/W}$

Example 4

Design the converter shown in figure given: $E_{unreg} = 9$ V dc, $V_{load} = 5$ V dc, $I_{dc\ load} = 0.8$ A dc, $f_{osc} = 150$ kHz, $\Delta v_{pp} = 0.1$ V

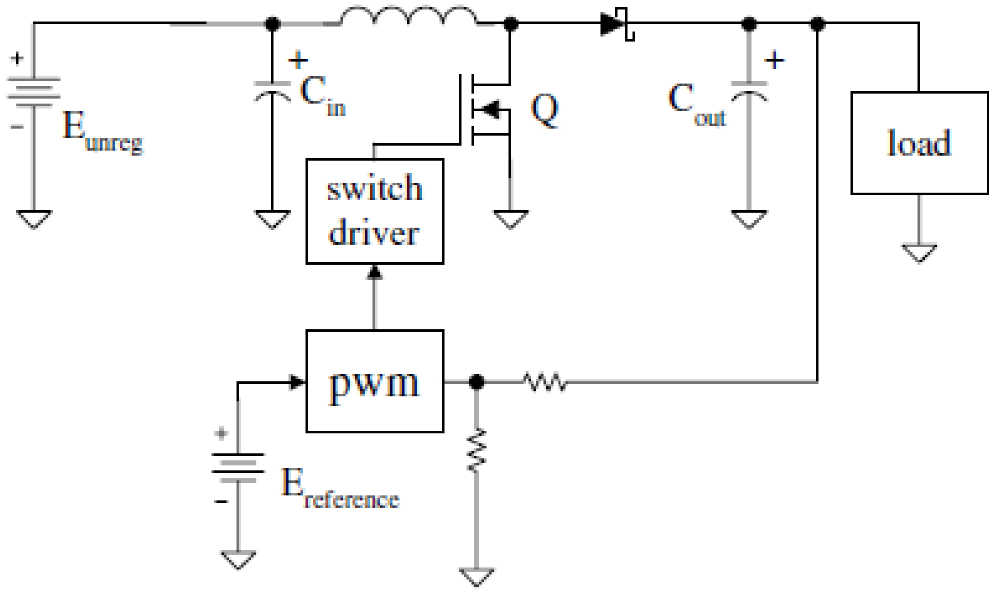
Find: $L, C_{out}, ESR_{Cout}, I_{dc\ in}, P_{unreg}$



Example 5

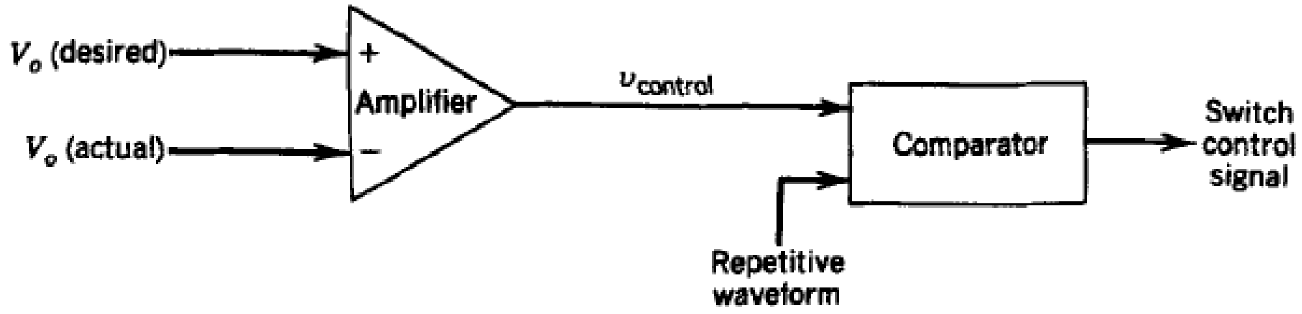
Design the converter shown in figure given: $E_{unreg} = 5$ V dc, $V_{load} = 9$ V dc, $I_{dc\ load} = 0.8$ A dc, $f_{osc} = 100$ kHz, $\Delta v_{pp} = 0.1$ V

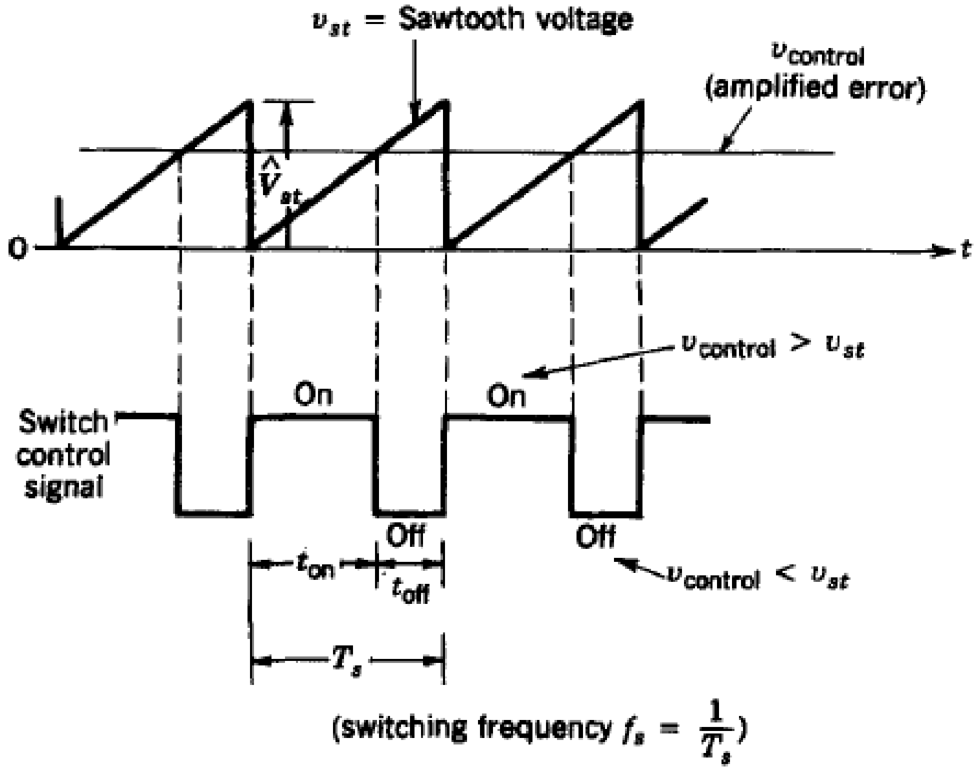
Find: $L, C_{out}, I_{dc\ in}, P_{unreg}$



PWM Working

Feedback the output voltage and compare it with the reference voltage.





Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan